Week 4 Lab Task

Vulnerability Analysis

Option B — Nmap Vulnerability Scripts (Quick & Lightweight)

Vulnerability Assessment Report:

Executive Summary

A vulnerability assessment was conducted on a simulated lab environment (Metasploitable 2). Multiple critical and high-risk vulnerabilities were identified across network services and web applications. These vulnerabilities could allow unauthorized access, data leakage, and remote code execution.

Target Information

- **Target IP:** 192.168.18.134
- **System:** Metasploitable 2
- **Operating System:** Linux (intentionally vulnerable)

Tools Used

- Nmap (vulnerability scanning)
- Nikto (web server scanning)

Vulnerability Findings

Critical Vulnerabilities:

Vulnerability	Severity	CVSS	Description	Fix
FTP Anonymous Login	Critical	9.8	Allows login without credentials	Disable anonymous FTP
vsftpd Backdoor	Critical	10.0	Remote command execution possible	Update/remove vulnerable service
SQL Injection (DVWA)	Critical	9.0	Database can be accessed/modified	Use input validation

High Vulnerabilities:

Vulnerability	Severity	CVSS	Description	Fix
Telnet Enabled	High	7.5	Sends credentials in plaintext	Disable Telnet, use SSH

Outdated Apache 2.2	High	7.5	Known exploits available	Upgrade Apache
SMB Null Session	High	7.2	Allows unauthenticated enumeration	Disable null sessions

Medium Vulnerabilities:

Vulnerability	Severity	CVSS	Description	Fix
Directory Listing Enabled	Medium	5.3	Files exposed publicly	Disable directory listing
Server Banner Disclosure	Medium	5.0	Reveals software version	Hide server headers

How Vulnerabilities Were Found

- `nmap --script=vuln` → detected known vulnerabilities

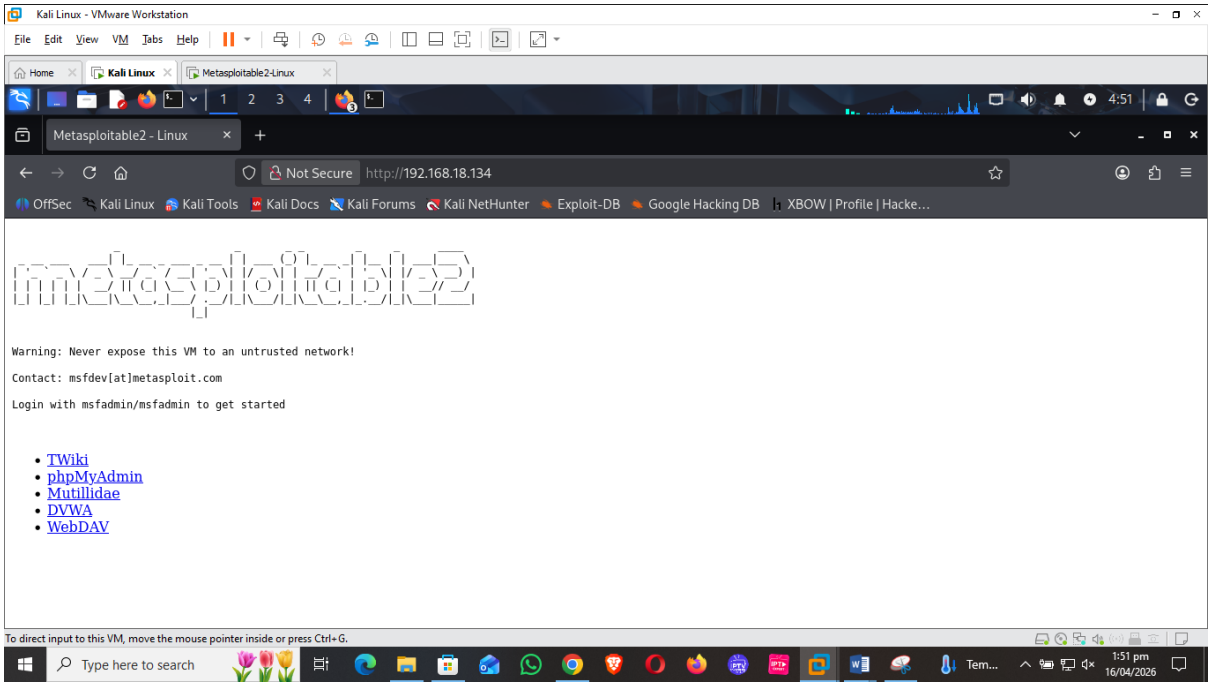
```
┌──(fahad㉿Kali)-[~]
└─$ nmap --script=vuln 192.168.18.134
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-16 04:31 EDT
Nmap scan report for 192.168.18.134
Host is up (0.0012s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
```

- `nmap --script=http-vuln*` → found web issues

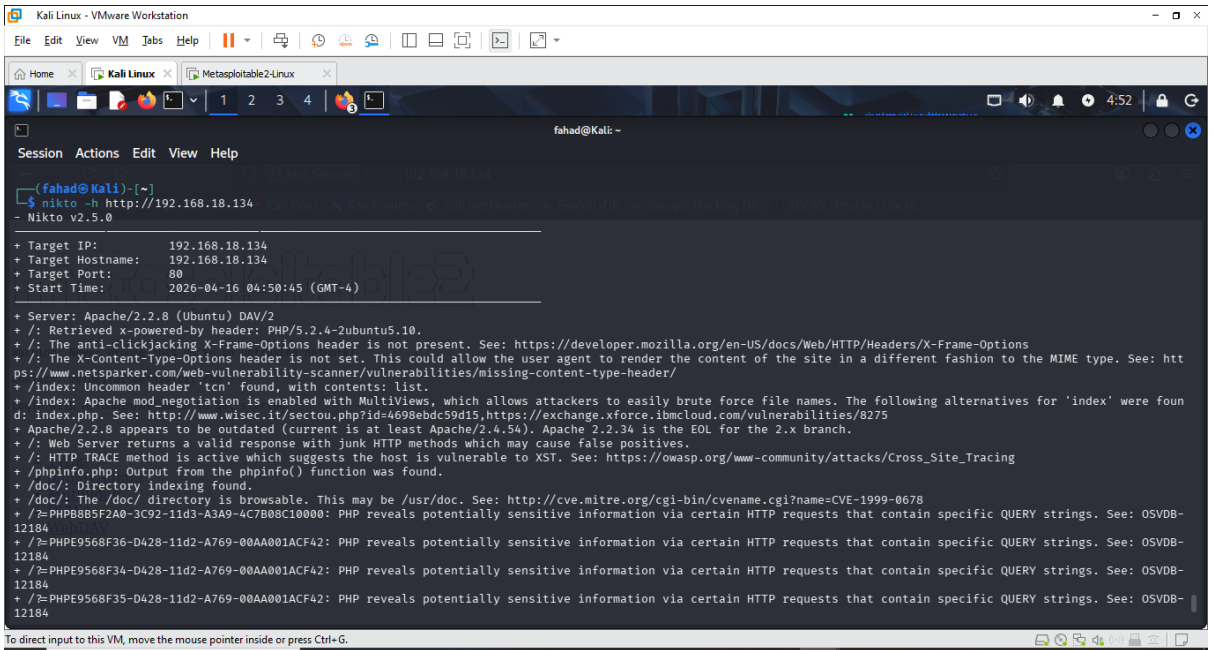
```
└─$ nmap --script=http-vuln* 192.168.18.134
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-16 04:41 EDT
Nmap scan report for 192.168.18.134
Host is up (0.0024s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
```

- `nmap --script=ftp-anon` → confirmed anonymous FTP access

```
┌──(fahad㉿Kali)-[~]
└─$ nmap --script=ftp-anon 192.168.18.134
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-16 04:45 EDT
Nmap scan report for 192.168.18.134
Host is up (0.0028s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp   open  ssh
```


Kali Linux - VMware Workstation
File Edit View VM Tabs Help
Home
Kali Linux
Metasploitable2-Linux
Metasploitable2 - Linux
Not Secure
http://192.168.18.134
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
XBOW | Profile | Hacke...
Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started
TWiki
phpMyAdmin
Mutillidae
DVWA
WebDAV
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.
Type here to search
1:51 pm
16/04/2026

- `nikto` → detected outdated Apache and misconfigurations



Risk Analysis (Prioritization)

Priority	Action
1	Fix Critical vulnerabilities immediately
2	Patch High-risk services
3	Address Medium issues gradually

Conclusion

The assessment revealed multiple exploitable vulnerabilities. If this were a real production system, it would be at high risk of compromise. Immediate remediation and continuous monitoring are strongly recommended.